Lab: Practice Project Guide

Estimated time needed: 1 hour

You have read the practice project overview and are now familiar with the scenario and tasks that need to be performed. The instructions comprising the step list for each task to be performed in the practice project are provided below.

Software used in this project

The instruction videos in this course use the full Excel Desktop version as this has all the available product features, but for the hands-on labs, we will be using the free "Excel for the web" version as this is available to everyone.

Although you can use the Excel Desktop software, it is recommended that you use Excel for the web for the project, as the instructions specifically refer to this version, and there are some minor differences in the interface and available features.

Dataset used in this project

The dataset used in this assignment is curated from the following source:
Baseball Salaries - dataset by natereed | data. world under a **Public Domain license**.

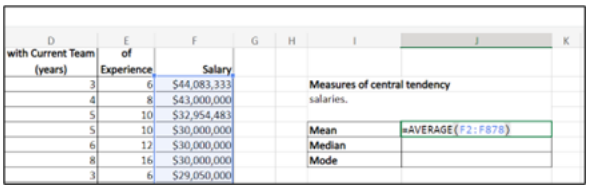
To be able to complete the project successfully, please use the data set provided with the project, rather than the dataset from the source because the dataset in the project has been modified from the source.

Download the Base_ball_player_salaries_Project_Start.xlsx file. To download it, right-click and open the link in a new tab. Then, open the file in Excel for the web.

Detailed guide to each task in the project

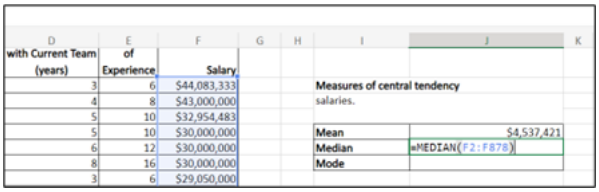
Task 1 - Measures of central tendency: Calculate the mean, median, and mode of the salaries. Perform the calculations in the cells provided.

1. Calculate the mean of salaries using the AVERAGE function. The range to be selected is the salaries column, F2 to F878. Calculate the mean in the cell provided for it.



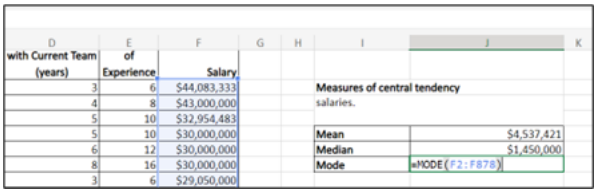
D	E	F	G	H	I	J	K
with Current Team (years)	of Experience	Salary					
3	6	$44,083,333			Measures of central tendency		
4	8	$43,000,000			salaries.		
5	10	$32,954,483					
5	10	$30,000,000			Mean	=AVERAGE(F2:F878)	
6	12	$30,000,000			Median		
8	16	$30,000,000			Mode		
3	6	$29,050,000					

2. Calculate the median of salaries using the MEDIAN function. The range to be selected is the salaries column, F2 to F878. Calculate the median in the cell provided for it.



D	E	F	G	H	I	J	K
with Current Team (years)	of Experience	Salary					
3	6	$44,083,333			Measures of central tendency		
4	8	$43,000,000			salaries.		
5	10	$32,954,483					
5	10	$30,000,000			Mean	$4,537,421	
6	12	$30,000,000			Median	=MEDIAN(F2:F878)	
8	16	$30,000,000			Mode		
3	6	$29,050,000					

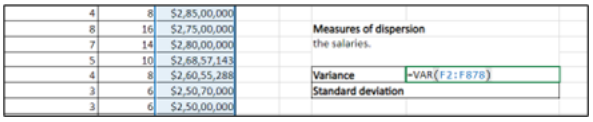
3. Calculate the mode of salaries using the MODE function. The range to be selected is the salaries column, F2 to F878. Calculate the mode in the cell provided for it.



D	E	F	G	H	I	J	K
with Current Team (years)	of Experience	Salary					
3	6	$44,083,333			Measures of central tendency		
4	8	$43,000,000			salaries.		
5	10	$32,954,483					
5	10	$30,000,000			Mean	$4,537,421	
6	12	$30,000,000			Median	$1,450,000	
8	16	$30,000,000			Mode	=MODE(F2:F878)	
3	6	$29,050,000					

Task 2 - Measures of dispersion: Calculate the variance and standard deviation for salaries. Perform the calculations in the cells provided.

1. Calculate the variance for salaries using the VAR function. The range to be selected is the salaries column, F2 to F878. Calculate the variance in the cell provided for it.



4	8	$2,85,00,000			
8	16	$2,75,00,000		Measures of dispersion	
7	14	$2,80,00,000		the salaries.	
5	10	$2,68,57,143			
4	8	$2,60,55,288		Variance	=VAR(F2:F878)
3	6	$2,50,70,000		Standard deviation	
3	6	$2,50,00,000			

2. Calculate the standard deviation for salaries using the STDEV.S function. The range to be selected is the salaries column, F2 to F878. Calculate the standard deviation in the cell provided for it.

4	8	$2,85,00,000			
8	16	$2,75,00,000		Measures of dispersion	
7	14	$2,80,00,000		salaries.	
5	10	$2,68,57,143			
4	8	$2,60,55,288		Variance	4.04822E+13
3	6	$2,50,70,000		Standard deviation	=STDEV.S(F2:F878)
3	6	$2,50,00,000			

Task 3 - Minimum and maximum: Find the minimum and maximum salary. Perform the calculations in the cells provided.

1. Calculate the minimum salary using the MIN function. The range to be selected is the salaries column, F2 to F878. Calculate the minimum salary in the cell provided for it.

2	4	$2,50,00,000			
5	10	$2,40,00,000		Minimum and maximum	
7	14	$2,40,00,000		Calculate the minimum salary and maximum salary	
3	6	$2,37,50,000			
7	14	$2,30,00,000		Minimum	=MIN(F2:F878)
7	14	$2,30,00,000		Maximum	
8	16	$2,27,50,000			

2. Calculate the maximum salary using the MAX function. The range to be selected is the salaries column, F2 to F878. Calculate the maximum salary in the cell provided for it.

2	4	$2,50,00,000			
5	10	$2,40,00,000		Minimum and maximum	
7	14	$2,40,00,000		Calculate the minimum salary and maximum salary	
3	6	$2,37,50,000			
7	14	$2,30,00,000		Minimum	$5,45,000
7	14	$2,30,00,000		Maximum	=MAX(F2:F878)
8	16	$2,27,50,000			

Task 4 – Quartiles and Percentiles: Calculate the following quartiles and percentiles in the cells provided for the calculation:

- First, second, and third quartiles of the salaries (Q1, Q2, Q3)
- 1, 5, 50, 75, 95, and 99 percentiles of the salaries (P01, P05, P50, P75, P95, P99)

1. Calculate the Quartiles using the QUARTILE function. The Quartile function has two parameters, "array" and "quart." The array is the salaries column, F2 to F878. The quart will be 1, 2,3, or 4, depending on the quartile which you are calculating.

4	8	$2,26,66,666			
5	10	$2,25,00,000		Quartiles and percentiles	
6	12	$2,23,57,142		Calculate the quartiles and percentiles of the salaries	
2	4	$2,21,77,778			
4	8	$2,21,42,857		Q1	=QUARTILE(F2:F878,1)
5	10	$2,20,00,000		Q2	
6	12	$2,20,00,000		Q3	

4	8	$2,26,66,666			
5	10	$2,25,00,000		Quartiles and percentiles	
6	12	$2,23,57,142		Calculate the quartiles and percentiles of the salaries	
2	4	$2,21,77,778			
4	8	$2,21,42,857		Q1	$5,57,471.00
5	10	$2,20,00,000		Q2	=QUARTILE(F2:F878,2)
6	12	$2,20,00,000		Q3	

4	8	$2,26,66,666			
5	10	$2,25,00,000		Quartiles and percentiles	
6	12	$2,23,57,142		Calculate the quartiles and percentiles of the salaries	
2	4	$2,21,77,778			
4	8	$2,21,42,857		Q1	$5,57,471.00
5	10	$2,20,00,000		Q2	$14,50,000.00
6	12	$2,20,00,000		Q3	=QUARTILE(F2:F878,3)

2. Calculate the Percentiles using the PERCENTILE function. The Percentile function has two parameters, "array" and "k." The array is the salaries column, F2 to F878. "k" is the percentile value between 0 and 1. For 1 percentile, the percentile value is 0.01. For the 5 percentile, the percentile value will be 0.05.

6	12	$2,20,00,000		Q3	$60,00,000.00
2	4	$2,20,00,000		P01	=PERCENTILE(F2:F878,0.01)
1	2	$2,18,57,142		P05	
8	16	$2,16,25,000		P50	
2	4	$2,16,25,000		P75	
5	10	$2,15,00,000		P95	
2	4	$2,13,59,375		P99	
8	16	$2,12,50,000			

6	12	$2,20,00,000		Q3	$60,00,000.00
2	4	$2,20,00,000		P01	$5,45,000.00
1	2	$2,18,57,142		P05	=PERCENTILE(F2:F878,0.05)
8	16	$2,16,25,000		P50	
2	4	$2,16,25,000		P75	
5	10	$2,15,00,000		P95	
2	4	$2,13,59,375		P99	
8	16	$2,12,50,000			

1	2	$2,18,57,142		P05	$5,45,320.00
8	16	$2,16,25,000		P50	=PERCENTILE(F2:F878,0.5)
2	4	$2,16,25,000		P75	
5	10	$2,15,00,000		P95	
2	4	$2,13,59,375		P99	
8	16	$2,12,50,000			

1	2	$2,18,57,142		P05	$5,45,320.00
8	16	$2,16,25,000		P50	$14,50,000.00
2	4	$2,16,25,000		P75	=PERCENTILE(F2:F878,0.75)
5	10	$2,15,00,000		P95	
2	4	$2,13,59,375		P99	
8	16	$2,12,50,000			

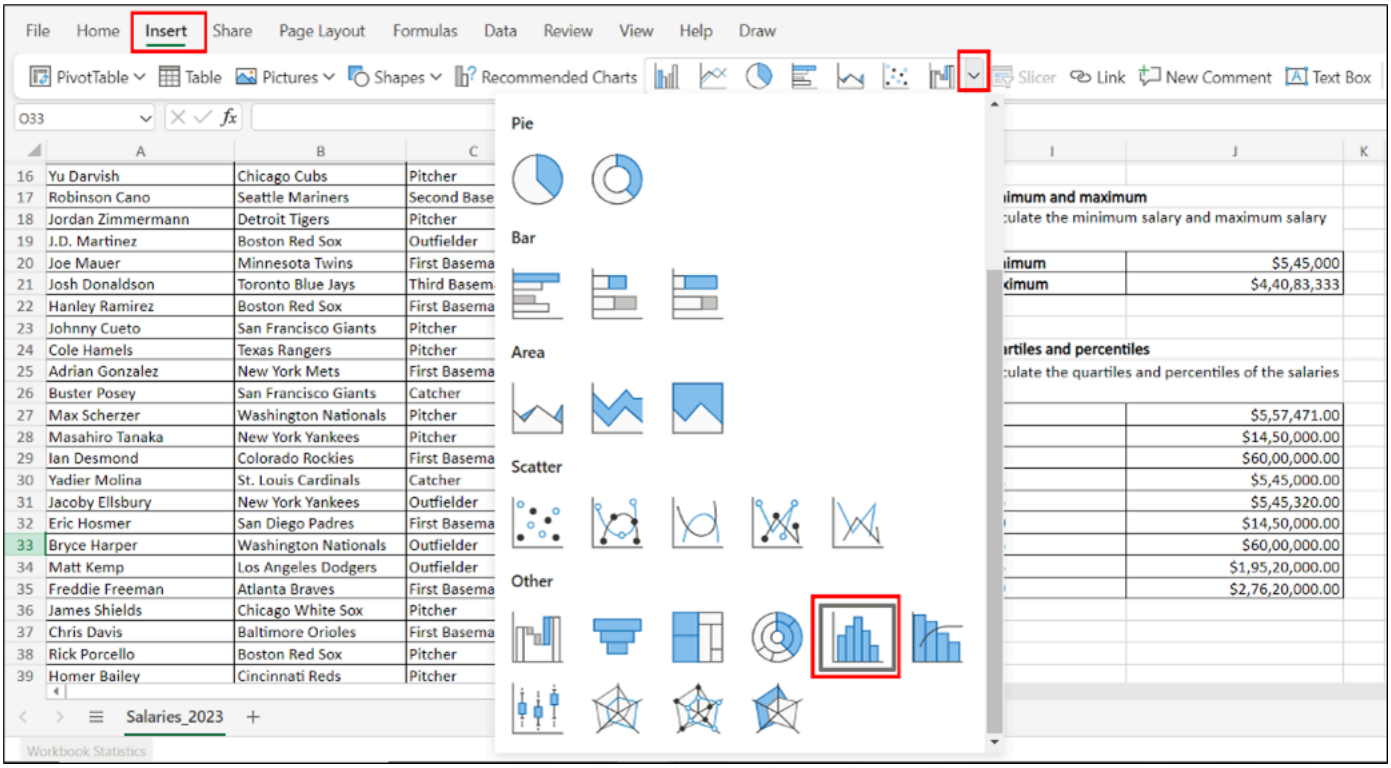
8	16	$2,16,25,000		P50	$14,50,000.00
2	4	$2,16,25,000		P75	$60,00,000.00
5	10	$2,15,00,000		P95	=PERCENTILE(F2:F878,0.95)
2	4	$2,13,59,375		P99	
8	16	$2,12,50,000			

8	16	$2,16,25,000		P50	$14,50,000.00
2	4	$2,16,25,000		P75	$60,00,000.00
5	10	$2,15,00,000		P95	$1,95,20,000.00
2	4	$2,13,59,375		P99	=PERCENTILE(F2:F878,0.99)
8	16	$2,12,50,000			

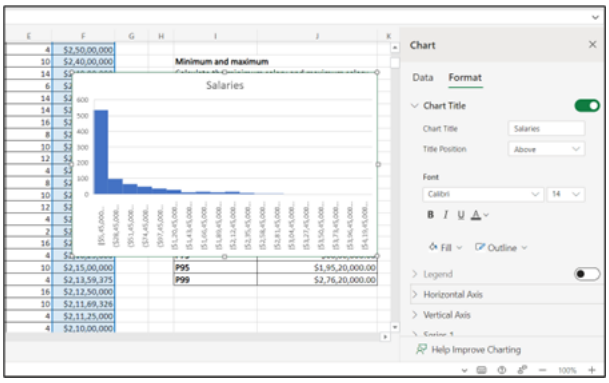
Task 5 – Histogram: Create a histogram showing the probability distribution of salaries. What does it tell you about the distribution of data?

To create a histogram:

1. Select the salary column.
2. Go to the Insert tab and select histogram under recommended charts. The histogram will be generated.



3. Click the chart to open the Data and Format panel to adjust how the chart is displayed.

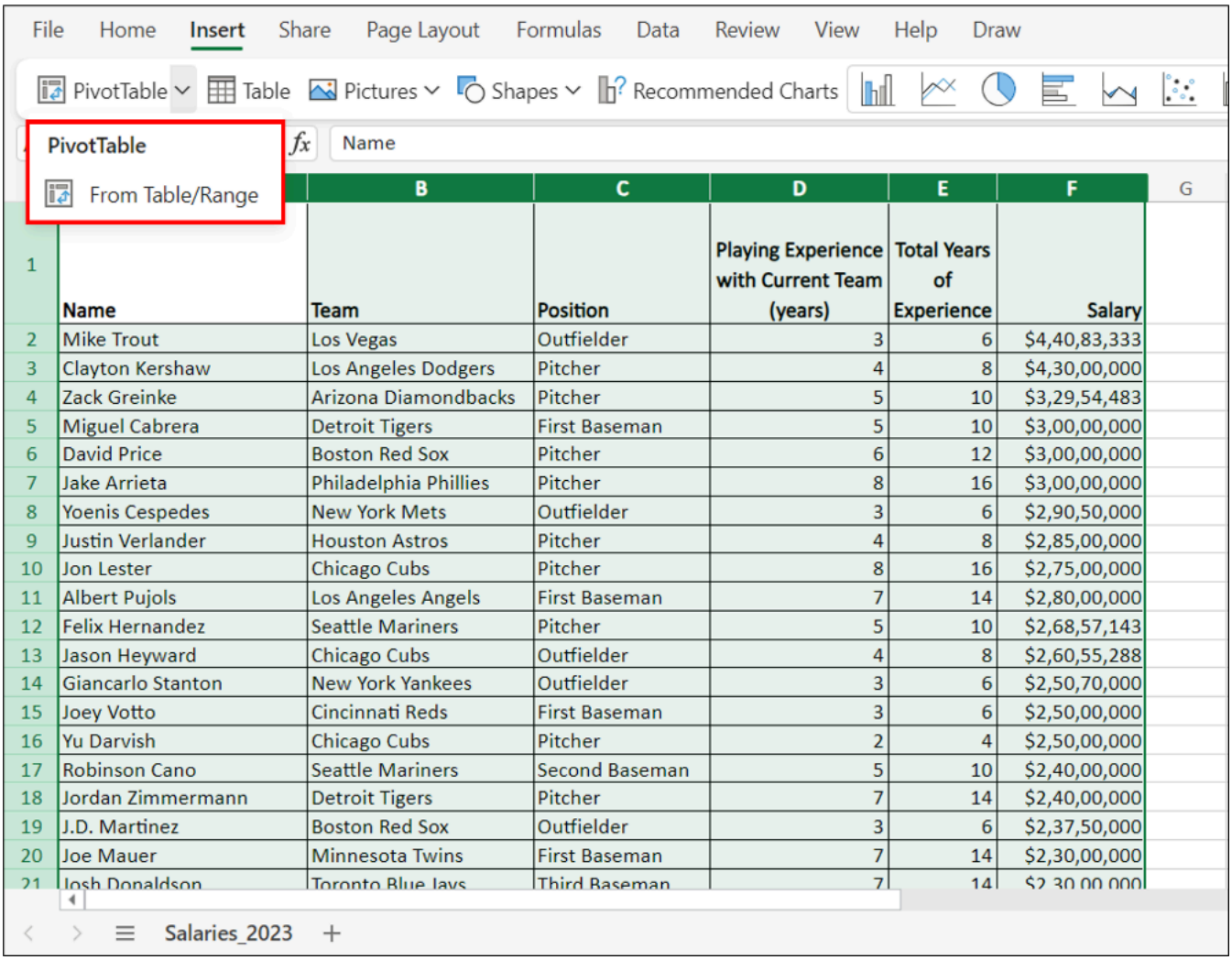


4. Interpretation: The distribution is skewed to the left.

Task 6a – Create a pivot chart to find the team paying the highest salaries to its players.

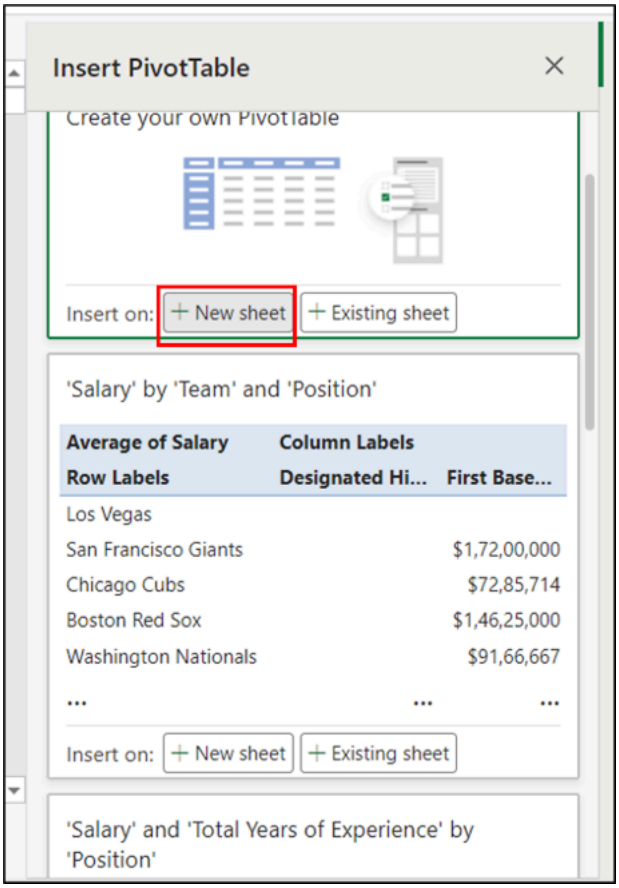
To create a pivot table:

1. Select the entire table.
2. Go to the Insert tab and click the arrow beside the Pivot table. Then, select From Table/Range.

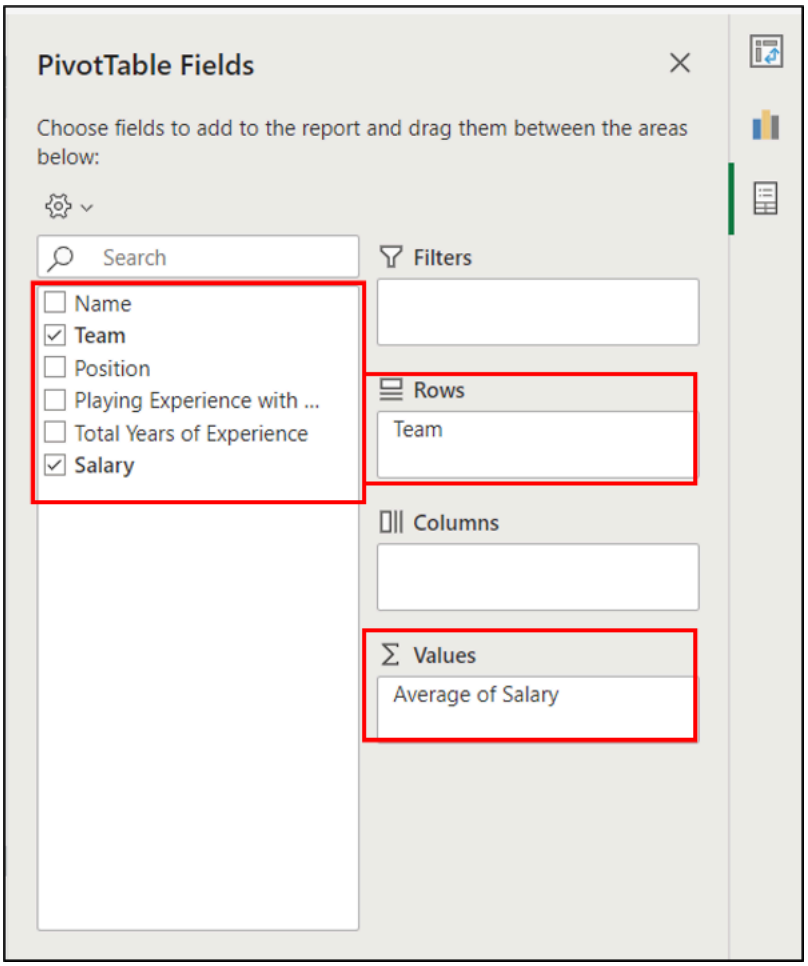


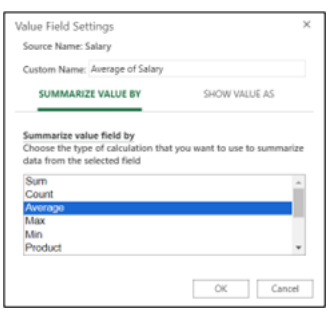
	Name	Team	Position	Playing Experience with Current Team (years)	Total Years of Experience	Salary
2	Mike Trout	Los Vegas	Outfielder	3	6	$4,40,83,333
3	Clayton Kershaw	Los Angeles Dodgers	Pitcher	4	8	$4,30,00,000
4	Zack Greinke	Arizona Diamondbacks	Pitcher	5	10	$3,29,54,483
5	Miguel Cabrera	Detroit Tigers	First Baseman	5	10	$3,00,00,000
6	David Price	Boston Red Sox	Pitcher	6	12	$3,00,00,000
7	Jake Arrieta	Philadelphia Phillies	Pitcher	8	16	$3,00,00,000
8	Yoenis Cespedes	New York Mets	Outfielder	3	6	$2,90,50,000
9	Justin Verlander	Houston Astros	Pitcher	4	8	$2,85,00,000
10	Jon Lester	Chicago Cubs	Pitcher	8	16	$2,75,00,000
11	Albert Pujols	Los Angeles Angels	First Baseman	7	14	$2,80,00,000
12	Felix Hernandez	Seattle Mariners	Pitcher	5	10	$2,68,57,143
13	Jason Heyward	Chicago Cubs	Outfielder	4	8	$2,60,55,288
14	Giancarlo Stanton	New York Yankees	Outfielder	3	6	$2,50,70,000
15	Joey Votto	Cincinnati Reds	First Baseman	3	6	$2,50,00,000
16	Yu Darvish	Chicago Cubs	Pitcher	2	4	$2,50,00,000
17	Robinson Cano	Seattle Mariners	Second Baseman	5	10	$2,40,00,000
18	Jordan Zimmermann	Detroit Tigers	Pitcher	7	14	$2,40,00,000
19	J.D. Martinez	Boston Red Sox	Outfielder	3	6	$2,37,50,000
20	Joe Mauer	Minnesota Twins	First Baseman	7	14	$2,30,00,000
21	Josh Donaldson	Toronto Blue Jays	Third Baseman	7	14	$2,30,00,000

3. The Insert PivotTable panel opens to the right. In the "Create your own Pivot table" section, select New sheet to open the pivot table in a new sheet.

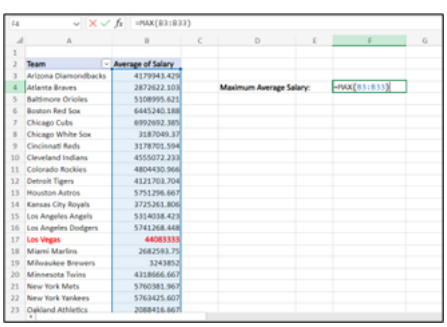


4. In the worksheet with the pivot table, you need to configure the settings in the right panel to get the required information. Since we need to find the highest-paying team, we need the average salary paid by each team. Therefore, in the Pivot-table panel, make the following settings.
   a. Select the columns Team and Salary.
   b. The Rows field shows Team.
   c. The Values field is set to Average of Salary. To do this:
      i. Select the dropdown button.
      ii. Select Value Field Settings.
      iii. In the Value Field Settings dialog box, in the Summarize Value By tab, select Average.





5. The teams and their average salaries are displayed. Find the maximum average salary using the MAX function. The team paying the highest salaries was Los Vegas.



Team	Average of Salary
Arizona Diamondbacks	4179943.429
Atlanta Braves	2872622.103
Baltimore Orioles	5108995.621
Boston Red Sox	6445240.188
Chicago Cubs	6992692.385
Chicago White Sox	3187049.37
Cincinnati Reds	3178701.594
Cleveland Indians	4555072.233
Colorado Rockies	4804430.966
Detroit Tigers	4121703.704
Houston Astros	5751296.667
Kansas City Royals	3725261.806
Los Angeles Angels	5314038.423
Los Angeles Dodgers	5741268.448
Los Vegas	44083333
Miami Marlins	2682593.75
Milwaukee Brewers	3243852
Minnesota Twins	4318666.667
New York Mets	5760381.967
New York Yankees	5763425.607
Oakland Athletics	2088416.667

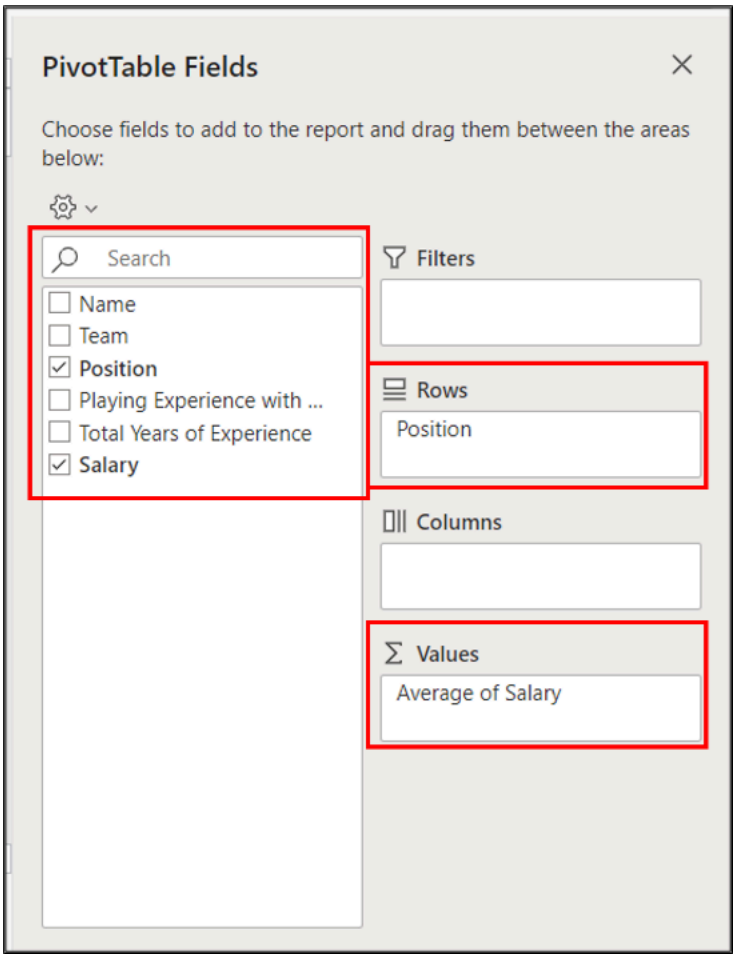
Maximum Average Salary: =MAX(B3:B33)

Task 6b – Find the position being paid the highest salary.

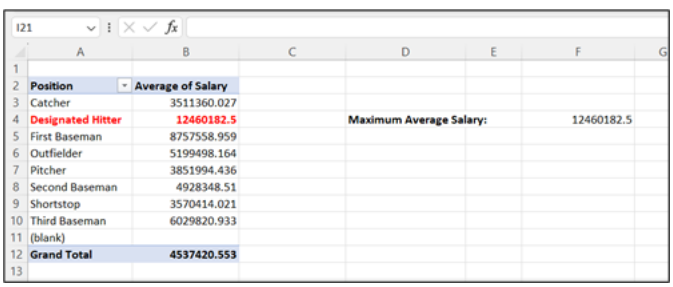
1. Create a pivot table following steps 1 to 3 in Task 6a. If the tool does not allow you to do so, duplicate the pivot table worksheet and re-configure the settings.

2. In the new pivot table sheet, configure the settings to get the required information. Since we need to find the highest-paid position, we need the average salary paid to each position. Therefore, in the Pivot-table panel, make the following settings.

   a. Select the Position and Salary checkboxes in the list of columns.
   b. The Rows field shows Position.
   c. The Values field is set to Average of Salary.



3. The positions and their average salaries are displayed. Find the maximum average salary using the MAX function. The position being paid the highest salaries is the Designated Hitter.



Position	Average of Salary
Catcher	3511360.027
Designated Hitter	12460182.5
First Baseman	8757558.959
Outfielder	5199498.164
Pitcher	3851994.436
Second Baseman	4928348.51
Shortstop	3570414.021
Third Baseman	6029820.933
(blank)	
Grand Total	4537420.553

Maximum Average Salary: 12460182.5

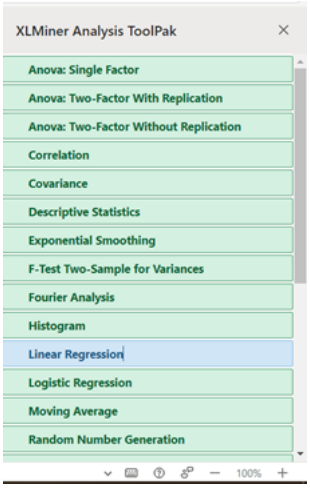
Task 7a – Perform a multiple linear regression analysis to predict the salaries for the next year. The salaries depend on the playing experience with the current team and the total years of experience.

1. Configure XL Miner
To conduct a regression analysis in Excel for the Web, you'll need to use an add-in. We recommend the XLMiner Analysis Toolpak which is a third-party add-in that is available for free in Excel for the Web. (In case you are using the desktop version of Excel, you may use the Data Analysis Toolpak available in the add-in.)

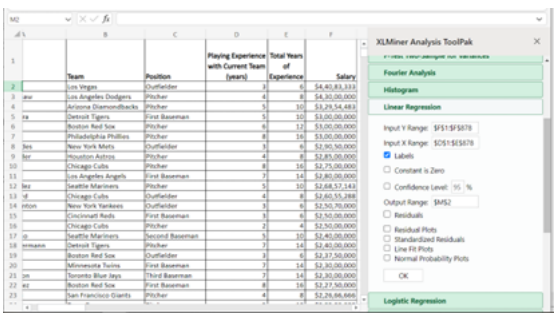
To configure the XLMiner Analysis Toolpak:
a. Click the File tab and then click Get Add-ins.
b. In the add-in store, search for XL Miner. The add-in will be listed. Click the Add button beside the add-in and accept the terms and conditions.

If you've already configured it previously, you'll find the add-in in My Add-ins tab in the Office Add-ins pop-up. Clicking on it will open XLMiner in a panel to the right of the screen.

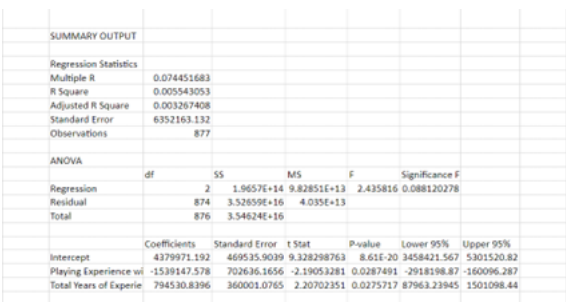
2. Run the regression analysis
a. Identify the dependent and independent variables. In this case, the dependent variable is the salary and the independent variables are the playing experience with current team and the total years of experience.
b. In the XLMiner Analysis Toolpak, click the Linear Regression tab.



c. Populate the inputs in the fields provided.
   i. Input Y Range: Salary (F1:F878).
   ii. Input X Range: Playing experience with the current team and Total years of experience (D1:E878).
   iii. If you've selected the column headers, select the labels checkbox.
   iv. Output Range: Select a cell on your worksheet from where the regression output will begin.
   v. Click OK.



d. The regression output will be displayed.



SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.074451683
R Square	0.005543053
Adjusted R Square	0.003267408
Standard Error	6352163.132
Observations	877

ANOVA

	df	SS	MS	F	Significance F
Regression	2	1.9657E+14	9.82851E+13	2.435816	0.088120278
Residual	874	3.52659E+16	4.035E+13		
Total	876	3.54624E+16			

	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%
Intercept	4379971.192	469535.9039	9.328298763	8.61E-20	3458421.567	5301520.82
Playing Experience wi	-1539147.578	702636.1656	-2.19053281	0.0287491	-2918198.87	-160096.287
Total Years of Experie	794530.8396	360001.0765	2.20702351	0.0275717	87963.23945	1501098.44

Task 7b – Interpret the output. Record the R-squared value and Significance F of the model and the p-values for each independent variable.

1. R-squared value = 0.00554
   This means that 0.5% of the variance in the salary is explained by the model.
2. Significance F = 0.08812
   This is the p-value of the model. The value is more than 0.05. So, the model is not statistically significant.
3. p-values
   The p-value of "Playing Experience with Current Team" is 0.0287 and the p-value of "Total Years of Experience" is 0.0276. The p-value of each independent variable is less than 0.05 showing that they individually have a statistically significant association with the salary.

SUMMARY OUTPUT						
Regression Statistics						
Multiple R	0.074451683					
R Square	0.005543053					
Adjusted R Square	0.003267408					
Standard Error	6352163.132					
Observations	877					
ANOVA						
	df	SS	MS	F	Significance F	
Regression	2	1.9657E+14	9.82851E+13	2.435816	0.088120278	
Residual	874	3.52659E+16	4.035E+13			
Total	876	3.54624E+16				
	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%
Intercept	4379971.192	469535.9039	9.328298763	8.61E-20	3458421.567	5301520.8
Playing Experience wi	-1539147.578	702636.1656	-2.19053281	0.0287491	-2918198.87	-160096.28
Total Years of Experie	794530.8396	360001.0765	2.20702351	0.0275717	87963.23945	1501098.4

Congratulations! You've completed the practice project. Now, you can move to completing the final graded project.

Author(s)

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